

control, root locus, Nyquist analysis

1 Control

- design of a servomechanism
- damped harmonic motion
- simulate to select parameter values

2 Root Locus

- moving poles

3 Nyquist analysis

- Nyquist plots
- the argument principle

MCS 472 Lecture 35
Industrial Math & Computation
Jan Verschelde, 10 April 2026

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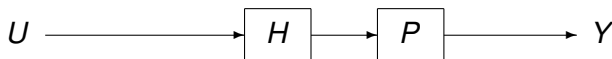
- moving poles

3 Nyquist analysis

- Nyquist plots
- the argument principle

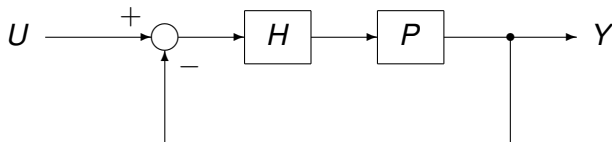
a control system with feedback

Consider a plant P , with controller H :



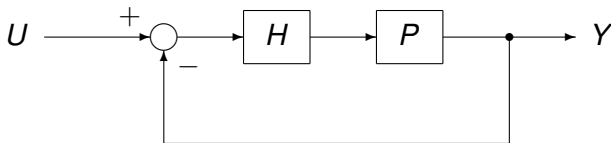
The above system is an open loop control system.

With feedback, we obtain a closed loop control system:



Example: steering control for an automobile.

design of a servomechanism



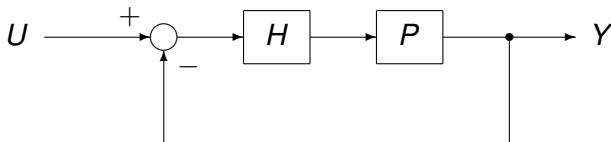
- The plant has transfer function $P(s) = 1/s$.
- The controller has transfer function $H(s) = \kappa + \alpha/s$ and is a *proportional integral* (PI) controller.

Problem:

Find the gains κ and α that yield a step response with

- at most a 20% overshoot, and
- a 1-second rise time, the time to reach one.

the transfer function of the closed loop system



- The open loop function has transfer function $H(s)P(s)$.
- The output is subtracted from the input in the closed loop:

$$Y(s) = H(s)P(s)(U(s) - Y(s)).$$

The transfer function of the closed loop system is then

$$(1 + H(s)P(s))Y(s) = H(s)P(s)U(s) \quad \Rightarrow \quad \frac{Y(s)}{U(s)} = \frac{H(s)P(s)}{1 + H(s)P(s)}.$$

the transfer function of the servomechanism

For the model of the servomechanism:

- The plant has transfer function $P(s) = 1/s$.
- The controller has transfer function $H(s) = \kappa + \alpha/s$.

The transfer function of the servomechanism is

$$G(s) = \frac{Y(s)}{U(s)} = \frac{H(s)P(s)}{1 + H(s)P(s)} = \frac{\kappa s + \alpha}{s^2 + \kappa s + \alpha}.$$

The response to the unit step is $U(s) = 1/s$.

$$Y(s) = G(s)U(s) = \left(\frac{\kappa s + \alpha}{s^2 + \kappa s + \alpha} \right) \frac{1}{s} = \frac{1}{s} - \frac{s}{s^2 + \kappa s + \alpha}$$

Exercise 1:

Apply the partial fraction decomposition to justify the last equality.

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damped harmonic motion

The Laplace transform of the output of the unit step is

$$Y(s) = \frac{1}{s} - \frac{s}{s^2 + \kappa s + \alpha}.$$

The denominator is quadratic of $Y(s)$,
which corresponds to a second order differential equation.

Substituting $\kappa = 2\zeta\omega_n$ and $\alpha = \omega_n^2$
introduces the standard notation for damped harmonic motion:

$$Y(s) = \frac{1}{s} - \frac{s}{s^2 + 2\zeta\omega_n s + \omega_n^2}.$$

rewriting the transfer function

$$Y(s) = \frac{1}{s} - \frac{s}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

is equivalent to

$$Y(s) = \frac{1}{s} - \frac{s + \zeta\omega_n}{D} + \left(\frac{\zeta}{\sqrt{1 - \zeta^2}} \right) \frac{\omega_d}{D},$$

with

$$D = (s + \zeta\omega_n)^2 + \omega_d^2, \quad \omega_d = \sqrt{1 - \zeta^2}\omega_n.$$

Notice the distinction between ω_n and ω_d .

the inverse Laplace transform

```
>>> from sympy import var
>>> from sympy import inverse_laplace_transform as ilap

>>> omega = var('omega', real=True)
>>> zeta = var('zeta', real=True)
>>> s, t = var('s, t')

>>> denominator = (s + zeta*omega)**2 + omega**2

>>> Y1 = - (s + zeta*omega)/denominator
>>> ilap(Y1, s, t)
-exp(-omega*t*zeta)*cos(omega*t)*Heaviside(t)

>>> Y2 = omega/denominator
>>> ilap(Y2, s, t)
exp(-omega*t*zeta)*sin(omega*t)*Heaviside(t)
```

the output in the time domain

The inverse Laplace transform of

$$Y(s) = \frac{1}{s} - \frac{s + \zeta\omega_n}{D} + \left(\frac{\zeta}{\sqrt{1 - \zeta^2}} \right) \frac{\omega_d}{D},$$

with

$$D = (s + \zeta\omega_n)^2 + \omega_d^2, \quad \omega_d = \sqrt{1 - \zeta^2},$$

is then

$$\begin{aligned} y(t) &= 1 - e^{-\zeta\omega_n t} \cos(\omega_d t) + \left(\frac{\zeta}{\sqrt{1 - \zeta^2}} \right) e^{-\zeta\omega_n t} \sin(\omega_d t) \\ &= 1 - e^{-\zeta\omega_n t} \left[\cos(\omega_d t) - \left(\frac{\zeta}{\sqrt{1 - \zeta^2}} \right) \sin(\omega_d t) \right]. \end{aligned}$$

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simulations to determine the overshoot

The ζ in

$$y(t) = 1 - e^{-\zeta\omega_n t} \left[\cos(\omega_d t) - \left(\frac{\zeta}{\sqrt{1-\zeta^2}} \right) \sin(\omega_d t) \right]$$

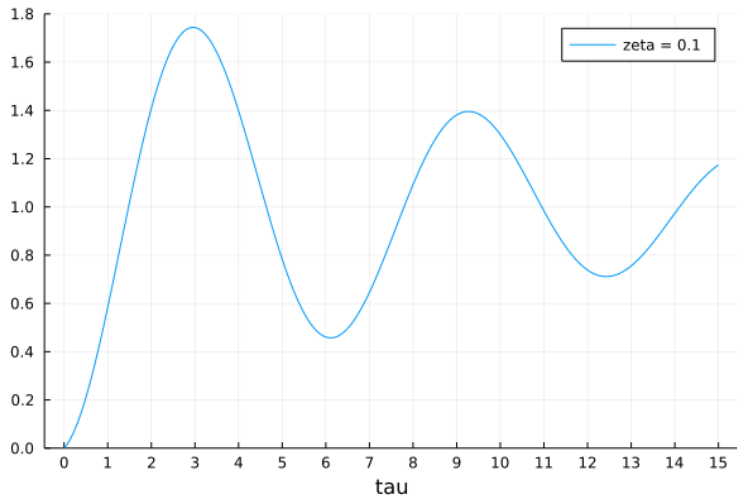
determines the overshoot, which should be less than 20%.

To study the effect of ζ , we go to normalized time

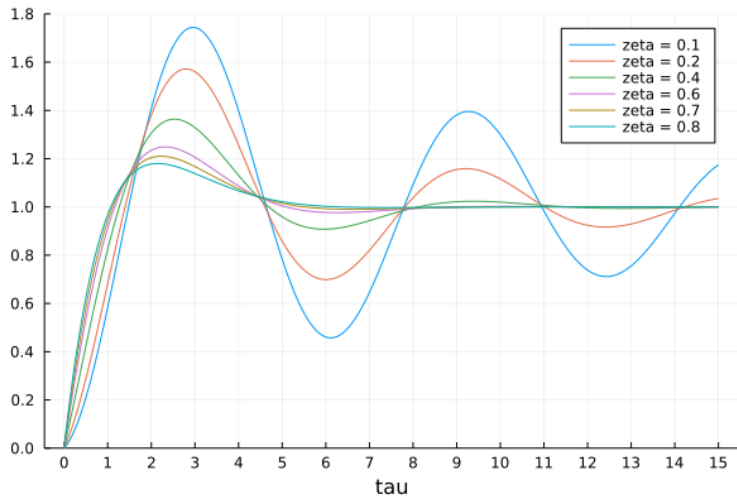
$$\tau = \omega_n t, \quad \omega_d = \sqrt{1-\zeta^2},$$

and graph $y(\tau)$, for various values of ζ .

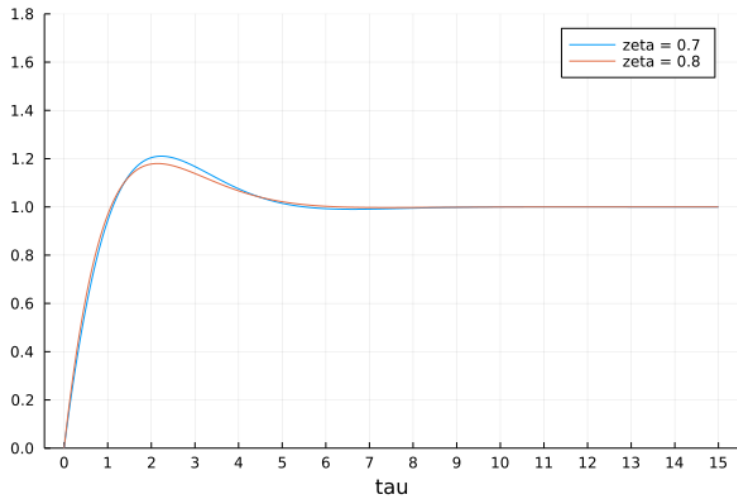
$$\zeta = 0.1$$



$$\zeta = 0.1, 0.2, 0.4, 0.6, 0.7, 0.8$$



$\zeta = 0.7, 0.8$



a solution to the design problem

- With $\zeta = 0.7$, we go slightly above 1.2,
so $\zeta = 0.8$ is a safer value to meet the less than 20% overshoot.
- With ω_n , we can set the rise time, as $\tau = \omega_n t$.
For $\zeta = 0.8$, at $\tau = 1$, we reach 0.97, and 1.01 at $\tau = 1.1$.
If we want one at $t = 1$, then choose $\omega = \tau/t = 1.1$.
- Use $\kappa = 2\zeta\omega_n = 1.76$ and $\alpha = \omega_n^2 = 1.21$
in the design of the controller.

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proportional feedback

Consider a plant with transfer function

$$P(s) = \frac{s^2 + 7s + 4}{s^3 - 3s^2 + 2s + 5}.$$

With proportional feedback $H(s) = \kappa$,
the closed loop system has transfer function

$$Q(s) = \frac{P}{1 + \kappa P}.$$

The denominator of $Q(s)$ is

$$\kappa (s^2 + 7s + 4) + s^3 - 3s^2 + 2s + 5.$$

The closed loop system is stable if and only if all roots of the denominator all lie in the open left plane $\text{real}(s) < 0$.

a Julia function, using SymPy

```
"""
```

```
function locus(kp::Float64, p)
```

returns the roots of the polynomial p in s defined by the coefficients where $kappa$ is replaced by kp .

```
"""
```

```
function locus(kp::Float64, p)
```

```
    cffs = [p.coeff(s,k) for k=0:3]
```

```
    c1 = [c.subs(Dict(kappa=>kp)) for c in cffs]
```

```
    cf1 = [Float64(c1[k]) for k=1:size(c1,1)]
```

```
    pls = sum([cf1[k]*s^(k-1) for k=1:size(cf1,1)])
```

```
    r1 = roots(pls)
```

```
    v1 = [Float64(real(r[1]))+Float64(imag(r[1]))*Complex(0,1)  
          for r in r1]
```

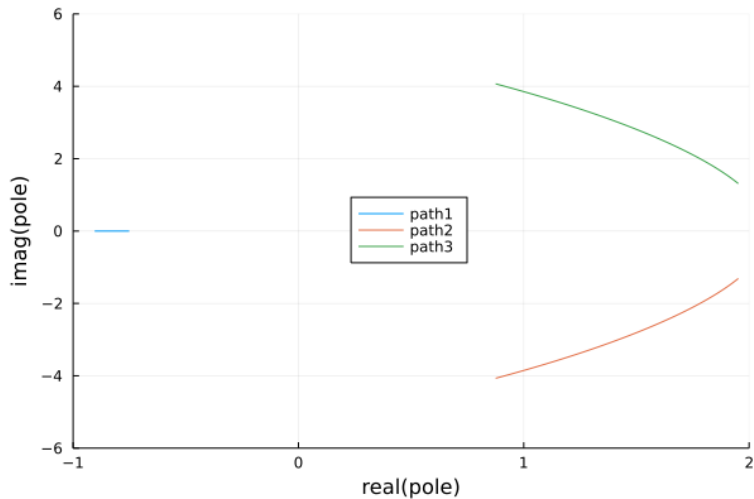
```
    tripletsort!(v1)
```

```
    return v1
```

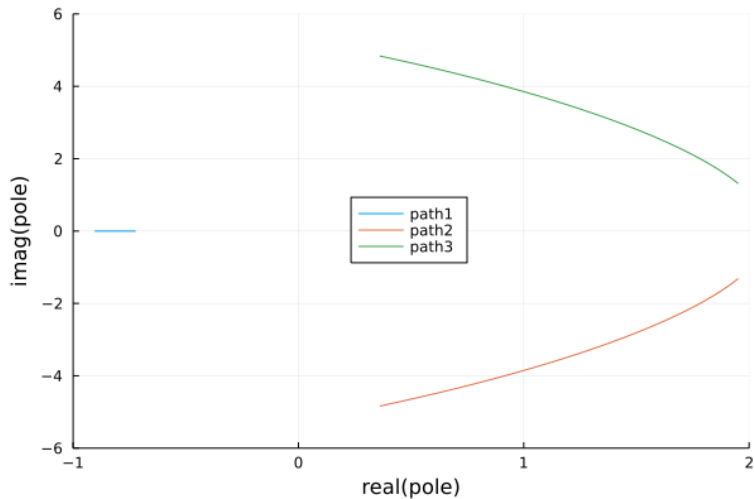
```
end
```

See the posted Jupyter notebook.

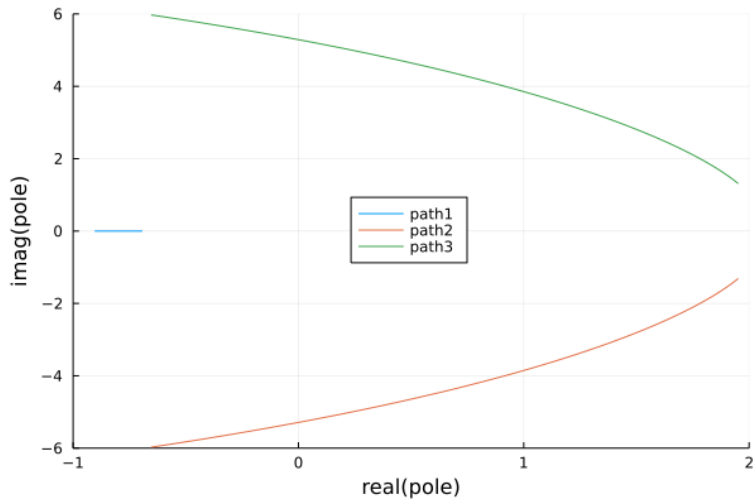
the poles for $\kappa = 2$, unstable



the poles for $\kappa = 3$, still unstable



the poles for $\kappa = 5$, stable



numerical continuation

Exercise 2:

For the previous example, find the critical threshold for κ , so the closed loop system is stable when κ crosses this threshold.

The function `locus` is inefficient and `tripletsort!` a hack.

Exercise 3:

A continuation method operates in two stages:

- Update κ as $\kappa + \Delta\kappa$, with step size $\Delta\kappa$.
- After each update, apply Newton's method to correct the root.

Apply this continuation method on the example on slide #19.

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Nyquist analysis

The stability of the closed loop system for proportional feedback is equivalent to finding a κ so that the roots of

$$\frac{1}{\kappa} + H(s)P(s) = 0$$

all lie in the open left half plane.

Nyquist analysis:

- 1 Set $G(s) = H(s)P(s)$.
- 2 Graph the image Γ of $z = G(s)$, for $s = \omega j$, for increasing ω .

All κ so that $-1/\kappa$ lies in the unbounded component of the z -plane disjointed by Γ will destabilize the closed loop system.

an example

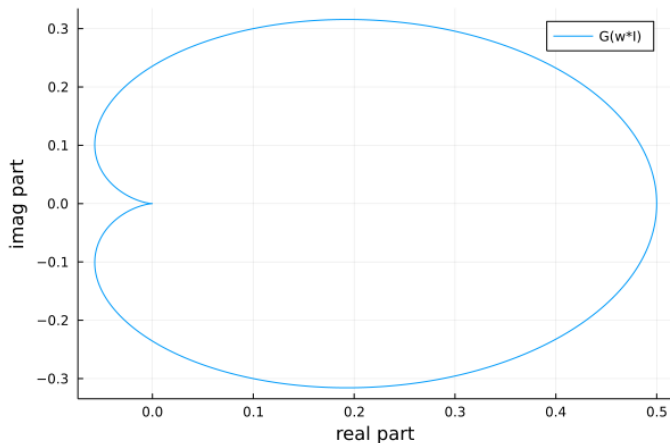
Consider:

$$P(s) = 1/(s - 1), \quad H(s) = 1/(s - 2), \quad G(s) = H(s)P(s).$$

Code to make the complex plot:

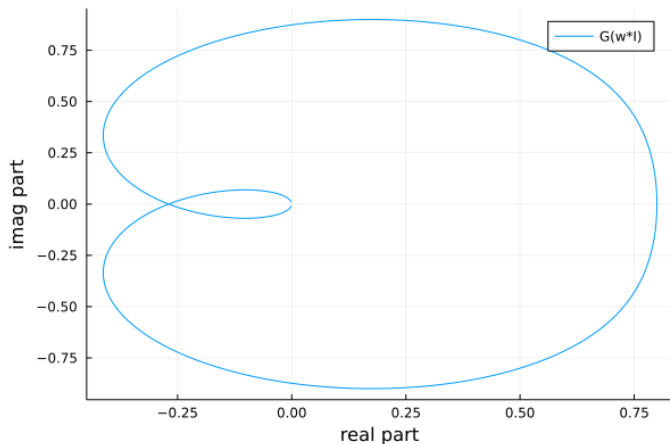
```
I = Complex(0, 1)
r(omega) = G(omega*I)
omegarange = -100:0.01:100
rt = [(real(r(t)), imag(r(t)))]
      for r in omegarange]
plot(rt, xlabel="real part",
      ylabel="imag part", label="G(w*I) ")
```

the complex plot of $1/((s-1)(s-2))$



The entire negative real axis is outside the curve,
thus there is no $\kappa > 0$ that stabilizes.

$$G(s) = (s^2 + 7s + 4)/(s^3 - 3s^2 + 2s + 5)$$



The entire negative real axis is outside the curve, except for $[-0.25, 0]$, so $-1/\kappa \in [-0.25, 0]$ or $\kappa > 4$, which agrees with the root locus method.

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the argument principle

Theorem (the argument principle)

- Let Ω be a domain in the complex plane.
- Suppose $f(s)$ is meromorphic (analytic everywhere, except for poles) on Ω .
- Let C be the closed curve in Ω that does not pass through any zeros or poles of $f(s)$.

Then:

$$\frac{1}{2\pi i} \int_C \frac{f'(s)}{f(s)} ds = \# \text{zeros} - \# \text{poles}$$

in Ω enclosed by C , counted with multiplicities.

As a corollary, **the winding number** $n(z_0, C)$ counts the number of roots minus the number of poles of $f(s) - z_0 = 0$ within C .

the Nyquist contour

Definition (the Nyquist contour)

For $R > 0$, *the Nyquist contour* is

- 1 the imaginary axis from $-Ri$ to $+Ri$, and
- 2 the semi-circle $Re^{i\theta}$, for θ from $+\pi/2$ to $-\pi/2$.

To check the stability of a rational plant $P(s) = N(s)/D(s)$,
check that no poles of P lie within the Nyquist contour for large R .

design an automobile cruise control system

Our text book has the design of an automobile cruise control system.

Proposal for a Topic of a Project:

Study the solution to this design problem in our text book.

Do the following:

- Describe the solution using your own words.
- Do the computational experiments to examine the stability.
- Write a report on your computational experiments.

root finding with the winding number

The winding number leads to a numerical method to approximate all roots of a general nonlinear equation in a domain of the complex plane.

Proposal for a Topic of a Project:

Study the proof of the argument principle theorem in our text book and make sure you understand the definition of the winding number.

Do the following:

- Describe the algorithm to compute all complex roots in a domain of a general nonlinear function.
- Experiment with the implementation in Pari/GP.
- Write a report on your computational experiments.

summary and bibliography

We introduced control with the example of a servomechanism, illustrating the notions of overshoot and rise time.

To examine the stability, apply root locus or Nyquist analysis.

We ended Chapter 10 of our text book.

- Charles R. MacCluer:
Industrial Mathematics. Modeling in Industry, Science, and Government. Prentice Hall, 2000.

An introductory text book in control gives many more examples:

- Richard C. Dorf and Robert H. Bishop:
Modern Control Systems. Ninth Edition, Prentice Hall, 2001.